**C 40604** **(Pages : 3)** **Name.....**

**Reg. No.....**

**SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023**

(CBCSS—UG)

Mathematics

MTS 6B 11—COMPLEX ANALYSIS

(2020 Admission onwards)

Time : Two Hours and a Half Maximum : 80 Marks

**Section A**

*Answer any number of questions.*
*Each question carries 2 marks.*
*Maximum 25 Marks.*

1. Define entire function. Give an example.
2. State a necessary condition for analyticity.
3. Prove that $u(x, y) = e^{-x} \sin y$ is harmonic.
4. Prove that $\overline{e^z} = e^{\bar{z}}$.
5. Find all values of $z$ satisfying the equation $e^{z-1} = -ie^3$.
6. Find the real and imaginary parts of $\sin(\bar{z})$.
7. Evaluate $\oint_C xydx + x^2\,dy$ where C is the curve $y = x^3, -1 \le x \le 2.$.
8. Define simply and multiply connected domains. Give examples for each.
9. State Cauchy's - Goursat theorem and find $\oint_C e^z\,dz$ on a simple closed contour C.
10. Evaluate the integral $\int_{\frac{i}{2}}^{i} e^{\pi z}\,dz$ and write it in the form $a + ib$.

**Turn over**

11. By using Cauchy's integral formula evaluate $\int_C \frac{z}{z^2+9}\,dz$ where C is the circle $|z-2i|=4$.

12. State root test.

13. Find the Taylor expansion of $f(z)=\frac{1}{1-z}$.

14. Find the Laurent's series expansion of $f(z)=\frac{\cos z}{z}$ in $0<|z|$.

15. Find the pole of $\frac{\sin z}{z^2}$.

## Section B

*Answer any number of questions.*
*Each question carries 5 marks.*
*Maximum 35 Marks.*

16. Prove that if $f$ is differentiable at a point $z_0$ in a domain D then $f$ is continuous at $z_0$

17. Find the real constants $a, b, c$ and $d$ so that $f(z)=(3x-y+5)+i(ax+by-3)$ is analytic.

18. Compute the principal value of the complex logarithm Ln $z$ for $z=i$ and $z=1+i$.

19. Find the derivative of the principal value of $z^i$ at the point $z=1+i$.

20. Find the upper bound of the absolute value of $\oint_C \frac{e^z}{z+1}\,dz$ where C is the circle $|z|=4$.

21. Evaluate $\oint_C \frac{1}{\sqrt{z}}\,dz$ where C is the line segment between $z_0=i$ and $z_1=9$.

22. Examine the covergence of the following series on their circle of convergence (a) $\sum_{0}^{\infty} z^n$ ; and

(b) $\sum_{0}^{\infty} \frac{z^n}{n^2}$.

23. Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for $|z| > 1$.

## Section C

*Answer any* **two** *questions.*
*Each question carries* 10 *marks.*
*Maximum* 20 *Marks*

24. (a) State and prove Cauchy's integral formula.

(b) Evaluate $\int_C \frac{z}{z^2+9} dz$ where C is the circle $|z-2i| = 4$.

25. Evaluate $\int_C \frac{dz}{z^2+1}$.

26. (a) State and prove Cauchy's inequality.

(b) State Maximum modulus theorem and find the maximum modulus of $f(z) = 2z + 5i$ on the closed circular region $|z| \leq 2$.

27. State and prove Cauchy's residue theorem and using it evaluate $\int_C \frac{dz}{z^3(z-1)}$ where C is $|z| = 2$.

**C 40604-A** **(Pages : 4)** **Name.....**

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**SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023**

(CBCSS—UG)

Mathematics

MTS 6B 11—COMPLEX ANALYSIS

(2020 Admission onwards)

(Multiple Choice Questions for SDE Candidates)

| **Time : 15 Minutes** | **Total No. of Questions : 20** | **Maximum : 20 Marks** |
|---|---|---|

## INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 11—COMPLEX ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Polar from of the Cauchy-Riemann equation is :

(A) $\frac{\partial u}{\partial r} = r\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -r\frac{\partial u}{\partial \theta}$.

(B) $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$.

(C) $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -r\frac{\partial u}{\partial \theta}$.

(D) $\frac{\partial u}{\partial r} = r\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$.

2. Which of the following function $f(z)$, of the complex variable $z$, is not analytic at all the points of the complex plane ?

(A) $f(z) = z^2$.

(B) $f(z) = e^z$.

(C) $f(z) = \sin z$.

(D) $f(z) = \log z$.

3. The function $f(z) = xy + iy$ is :

(A) Nowhere analytic.

(B) Analytic every where.

(C) Analytic only at origin.

(D) Analytic except at the origin.

4. If $z$ is a non zero complex number, then for $n = 1, 2, 3, \ldots$, $z^{1/n}$ is :

(A) $\exp(n \log z)$.

(B) $\exp\left(\frac{1}{n}\log z\right)$.

(C) $\exp\left(\frac{1}{n}\log\frac{1}{z}\right)$.

(D) $\exp\left(n \log\frac{1}{z}\right)$.

5. $|\sin az|^2 =$

(A) $\sin^2 ax + \sinh^2 ay$.

(B) $\sin^2 ax - \sinh^2 ay$.

(C) $\cos^2 ax + \sinh^2 ay$.

(D) $\cos^2 ax + \sinh^2 ay$.

6. The integral of $\oint_C \frac{dz}{z^2+9}$, where C is the unit circle is :

(A) 0.

(B) 1.

(C) 3.

(D) – 3.

7. The integral of the function $\int_C \frac{\cos z}{2z}\,dz$ where C is the unit circle is :

(A) $\pi$.

(B) $-\pi i$.

(C) $\pi i$.

(D) $2\pi i$.

8. The integral of the function $\int_C \frac{dz}{3z^2+1} dz$ where C is the circle $|z| = 1$ is :

(A) $\pi$. (B) $-\pi$.

(C) 0. (D) 1.

9. If $f$ is continuous in a domain D and if $\int_C f(z)\, dz = 0$ for every simple closed positively oriented contour C in D, then :

(A) $f$ is analytic in D. (B) $f$ is real valued in D.

(C) $f$ is constant in D. (D) $f$ is imaginary in D.

10. Piecewise smooth curve is also known as :

(A) Contour. (B) Smooth curve.

(C) Circle. (D) Regular curve.

11. The radius of convergence of the series $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ is :

(A) 1. (B) 0.

(C) $-1$. (D) $\infty$.

12. The center of the power series $\sum_{n=0}^{\infty} (z+4i)^n$ is :

(A) $4i$. (B) $2i$.

(C) 4. (D) $-4i$.

13. The zero of the function $\frac{z}{\cos z}$ is :

(A) 1. (B) 0.

(C) $-1$. (D) $\pi$.

14. The principal part of $f(z)$ at $z_0$ consists of infinite number of terms, then $z_0$ is known as :

(A) Pole. (B) Essential singular point.

(C) Removable singular point. (D) Simple pole.

15. The power series $b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots.$ converges

(A) Inside of some circle $|z| = R$. (B) On the circle $|z| = 1$.

(C) On some circle $|z| = R$. (D) Outside of some circle $|z| = R$.

16. Residue at $z = 2$ of the function $\frac{2z+1}{z^2 - z - 2}$ is :

(A) $\frac{5}{3}$. (B) $\frac{1}{3}$.

(C) $\frac{3}{5}$. (D) $\frac{2}{3}$.

17. Residue of the function $\frac{1}{z + z^2}$ at $z = 0$ is :

(A) 0. (B) 1.

(C) – 1. (D) 2.

18. Residue of the function $\frac{\sin z}{z^4}$ is :

(A) $\frac{2}{5}$. (B) $\frac{3}{5}$.

(C) $\frac{1}{6}$. (D) $-\frac{1}{6}$.

19. Given $f(z) = \frac{z^2}{z^2 + a^2}$. Then :

(A) $z = ia$ is a pole and $\frac{ia}{2}$ is a residue at $z = ia$ of $f(z)$.

(B) $z = ia$ is a simple pole and $ia$ is a residue at $z = ia$ of $f(z)$.

(C) $z = ia$ is a simple pole and $-\frac{ia}{2}$ is a residue at $z = ia$ of $f(z)$.

(D) None of the above.

20. Integration of $\int_C \frac{dz}{z \sin z}$, where C is $x^2 + y^2 = 1$ is :

(A) 1. (B) 2.

(C) 0. (D) – 1.